CEARF-N: Bounded Out-of-Fit Dynamic Rank Allocation for Sparse Next-Item Recommendation

Pham Quoc Cuong[1,2], Nguyen Ngoc Minh[1,2], Le Quang Minh[1,2], and Nguyen Hoai Nguyet An[1,2]

[1] University of Information Technology, Ho Chi Minh City, Vietnam
[2] Vietnam National University Ho Chi Minh City, Vietnam
{cuongpq.20,minhnn.20,minhlq.20,annhv.20}@grad.uit.edu.vn

Abstract. Explicit memory and neural rankers fail on different next-item queries, while rank-level hybrids commonly assign every query one global fusion weight. We introduce CEARF-N (*Contextual Expert-Allocation Rank Fusion with a Neural residual*). It is a constrained post-hoc conditional allocator over frozen rankers: a global prior and bounded query correction are learned continuously from source-disjoint out-of-fit training ranks. A four-parameter gate predicts $\beta_q = \beta_{\mathrm{OOF}} + \Delta_{\mathrm{eff}} \tanh(\mathbf{w}^\top \mathbf{z}_q + b)$, $\Delta_{\mathrm{eff}} \leq .10$, from three standardized, target-free inference features. No validation/test label fits or selects a mixing weight. Its expert interface is rank-only, making it invariant to strictly monotone score transforms, and it has a closed-form pair-intervention bound. Full-catalogue, three-seed experiments on two Amazon domains and Diginetica show that equal/dynamic fusion improves R@20 by 4.6–17.2% over the stronger constituent, confirming expert complementarity. The dynamic-specific finding is narrower: against the training-only global policy, dynamic allocation has interval-above-zero early-rank gains on Video, while Baby and Diginetica shifts are undetectable. A fixed same-β-multiset reassignment is worse on Video nDCG@20. A selection-procedure audit shows that one training-learned global scalar with zero validation cells comes within 2.5×10^{-4} R@20 of a fifteen-cell oracle on every domain, while endpoint-only selection loses .008–.023; replacing the neural expert with a stronger NARM preserves the fusion gain (+.0076 R@20 on Diginetica), and matched-teacher comparisons attribute most of the Amazon gain to metadata rather than the fusion mechanism.

Keywords: next-item recommendation · dynamic rank fusion · bounded rank allocation · sparse recommendation

1 Introduction

Modern next-item models strengthen learned representations through recurrent, state-space, semantic, and mixture-of-experts designs [10,12,11,6,15]. Sparse catalogues expose their complement: a rare item may receive few gradient updates while an observed transition or near-duplicate session remains decisive. Explicit memory cannot generalize beyond observed edges and overlap. The practical question is therefore how much each query should trust heterogeneous evidence.

A global fusion coefficient assigns every query the same trust. A flexible router adds another estimation surface; separately, benchmark shortcuts and weak tuning can inflate apparent progress [5,8]. We instead learn a continuous global prior from out-of-fit training ranks and permit only a bounded query correction. CEARF-N pairs PASGR, a compact semantic GRU, with a memory expert selected from a declared transition, neighbour-session, and popularity family; the training-only lock selects neighbour-session memory in every reported domain. The pair is combined by weighted reciprocal-rank fusion [1]. Validation may audit the frozen allocator, and an upstream study had already frozen PASGR, but neither role fits or selects β. The proposed object is constrained post-hoc conditional stacking over frozen rankers, not a new sequence encoder: rank-space removes raw-score scale, source-disjoint OOF fitting avoids expert self-fit targets, and a residual bound limits every query-wise intervention.

We ask how much gain is endpoint complementarity versus learned allocation (RQ1), whether a bounded residual adds conditional value beyond OOF-global allocation (RQ2), how robust it is to capacity/operator changes (RQ3), and what inference cost it adds (RQ4). Two auditing questions complete the analysis: how the full pipeline compares against simpler selection procedures under explicit tuning budgets (RQ5), and what matched-information and expert-swap controls isolate (RQ6).

Our contributions are:

1. a continuous training-only objective that learns both a global fusion prior and a bounded per-query correction from reciprocal-rank pairwise evidence, without enumerating β;
2. a fully specified rank-interface allocator with five optimized coefficients—one OOF global logit plus a four-parameter residual gate using only context length, last-item frequency, and a tail indicator;
3. a three-level attribution protocol separating endpoint complementarity, OOF-global allocation, and query assignment, with capacity and operator controls; and
4. a serving audit decomposing feature+gate, rank-fusion, and expert retrieval latency; and
5. a selection-procedure audit with explicit tuning budgets, plus matched-information and expert-swap controls that separate metadata gains from mechanism gains.

The claim is not that routing always wins, but that a small auditable controller recovers a reproducible early-rank allocation signal on Video.

2 Related Work

Experts and fusion. Recent session recommenders use state-space recurrence, MoE routing, semantic initialization, and long-tail constraints [12,11,6,15]. V-

SKNN/STAN retrieve related sessions [13,3]; NARM is an attentive recurrent reference [9]. RRF combines incommensurate score spaces through ranks [1]. The primitive is established; our contribution is an OOF-trained rank prior plus a raw-score-free residual with target-free inference features, frozen experts, and an explicit intervention bound.

Conditional allocation and evaluation. Neural MoE recommenders route jointly trained representations [11]; multi-channel retrieval can learn query-dependent ranking over channel signals [4]. We instead study a five-scalar bounded composition over frozen explicit and neural rankers. Because shortcut-solvable benchmarks and weak offline protocols can inflate progress [5,8], we report tuned transition/V-SKNN/STAN baselines, both fusion endpoints, matched-query intervals, operator controls, and semantic-teacher provenance. A held-out validation gate would consume validation as meta-training; our source-disjoint OOF fitting keeps declared validation labels out of allocation. The stacking lineage is detailed in the supplement.

3 CEARF-N

3.1 Expert interface: memory and neural ranks

For query $q = (i_1, \ldots, i_L)$, the memory expert exposes three top-120 candidate memories: distance/recency-weighted four-step transitions, IDF/recency-weighted neighbour sessions, and global popularity. Six fixed rank-weight profiles are evaluated on a disjoint training-only lock subset. The realized primary profile is session-only, $(0, 1, 0)$, in every domain and length regime; consequently its agreement bonus is inactive, while transition and popularity remain controls.

The selected profile first produces one memory ordering π_M. Equation 1 is then applied afresh to π_M and the neural ordering π_N, so the final expert evidences lie in $[0, 1]$. If ordering e assigns rank $r_e(x)$ to item x, its evidence is

$$\rho_e(x) = \frac{k+1}{k + r_e(x)}, \qquad k = 20, \tag{1}$$

with zero evidence when x is absent. PASGR supplies a separate neural top-120 list; in our locked cells it is a semantic GRU selected from the *Prototype-Aligned Semantic Graph Retrieval* candidate family. It encodes the prefix and scores the full catalogue by normalized state–item dot product. Its cached semantic teacher is documented as E5-small on Video [14] and TF–IDF/SVD otherwise; the frozen upstream cell is detailed in the supplement. Prototype transport is off in all three locked cells.

3.2 Continuous query-conditioned allocation

Let ρ_M and ρ_N denote the selected memory and neural evidence. The final score is

$$F_q(x) = (1 - \beta_q)\rho_M(x) + \beta_q \rho_N(x). \tag{2}$$

First, a scalar $\beta_{\mathrm{OOF}} \in (0,1)$ is learned continuously on out-of-fit training queries. Then the primary gate predicts

$$\beta_q = \beta_{\mathrm{OOF}} + \Delta_{\mathrm{eff}} \tanh(\mathbf{w}^\top \widetilde{\mathbf{z}}_q + b), \quad \Delta_{\mathrm{eff}} = \min\{\Delta, \beta_{\mathrm{OOF}}, 1-\beta_{\mathrm{OOF}}\},\ \Delta = .10, \tag{3}$$

where $\widetilde{\mathbf{z}}_q$ is standardized using training-only statistics. The three target-free features are $\log(1+L)$, $\log(1+\mathrm{freq}(i_L))$, and whether i_L lies outside the smallest most-popular item set covering 80% of training interactions. They are available before expert scoring; rank-overlap diagnostics are reserved for capacity ablations. The gate is linear inside the bounded residual; hence it adds four learned parameters, not another sequence model. With $\delta_q = \beta_q - \beta_{\mathrm{OOF}}$ and $D_q(x) = \rho_N(x) - \rho_M(x)$, a pair's global-fusion order is certified whenever $|m_{\mathrm{OOF}}(a,b)| > |\delta_q||D_q(a) - D_q(b)|$ (coarsely $> 2\Delta_{\mathrm{eff}}$). The gate is therefore a bounded expert-disagreement correction. Because experts enter only through ranks, strictly monotone raw-score transforms preserving each top-120 ordering leave the allocator and final ranking unchanged. For target y_q and negative x, the signed advantage $A_{qx} = [\rho_N(y_q) - \rho_N(x)] - [\rho_M(y_q) - \rho_M(x)]$ is both the derivative of the target margin with respect to β_q and the sign-determining term in the pairwise learning signal.

For calibration query q with target y_q, let $\mathcal{H}_q$ contain up to 32 highest-evidence non-target candidates from the union of both expert lists. We minimize

$$\mathcal{L} = \frac{1}{|\mathcal{Q}|}\sum_q \frac{1}{|\mathcal{H}_q|}\sum_{x\in\mathcal{H}_q} \mathrm{softplus}\left(\frac{F_q(x) - F_q(y_q)}{\tau}\right) + \lambda_a \overline{\beta_q} + \lambda_p \overline{(\beta_q - \beta_{\mathrm{OOF}})^2}, \tag{4}$$

with $\tau = .10$ and $\lambda_a = \lambda_p = 10^{-3}$. Queries whose target is absent from both top-120 lists are excluded because no β_q can retrieve the missing target from their union. The first regularizer weakly biases allocation toward lower neural mass unless opposed by the pairwise rank loss; the second shrinks query corrections toward the global policy. Increasing β_q decreases a pair's rank-loss term exactly when the neural target-versus-negative margin exceeds the memory margin. Stage 1 minimizes rank loss plus $\lambda_a \beta$ for 80 epochs; Stage 2 freezes the scalar and fits only $(\mathbf{w}, b)$ under Eq. 4 for 50 epochs with AdamW weight decay 10^{-4}. No candidate set of β values exists.

3.3 Leakage-safe training protocol

Before modeling, a stable hash declares 5,000 validation queries; their sources are ineligible for OOF calibration. A second hash removes at most 5,000 remaining training source sessions before expert fitting: 1,000 lock the memory profile and 4,000 produce the ranks used in Eqs. 3–4. Thus profile and gate subsets are disjoint, and neither source session contributes events to the index or PASGR fit that predicts its OOF target. Diginetica validation target events and incoming transitions are removed from tuning histories. This is a single OOF holdout, not full K-fold cross-fitting; allocator parameters remain frozen for final test.

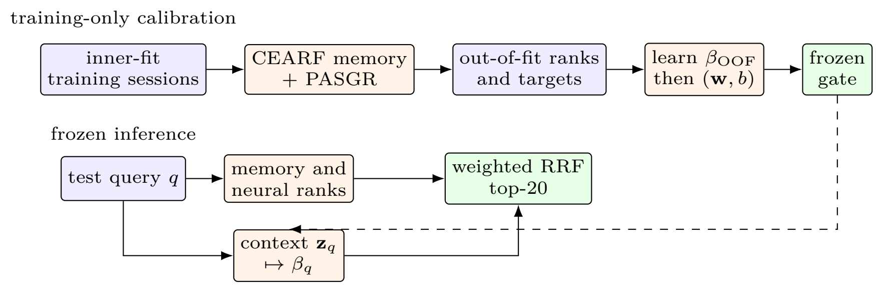

Fig. 1. CEARF-N learns allocation from held-out *training* sources, freezes the gate, and then performs query-conditioned rank fusion. Declared-validation/test labels never fit or select allocator parameters.

4 Experimental Setup

We derive 5-core leave-last-out histories from Amazon Reviews 2023 for Video Games and Baby Products [7]; previously seen items are masked. Diginetica uses its public session split and permits repeated items [2]. The domains contain respectively 25,613/36,014/43,098 items, 94,762/150,777/186,670 training sequences, and 94,762/150,777/60,858 test queries; 45% of Diginetica queries but none of the Amazon queries have $L \leq 2$. We use "sparse" for item-frequency sparsity: 27.9%, 27.9%, and 19.3% of train-catalogue items occur at most five times in Video, Baby, and Diginetica. No sampled negatives are used at evaluation. We report Recall at 6, 10, and 20, nDCG at the same cutoffs, and $U = .5\,\mathrm{R@6} + .5\,\mathrm{R@20}$; R@10 and all nDCG cutoffs guard interpretation against that utility. Stochastic runs use seeds 42, 123, and 456. Main-table comparators are transition-only, V-SKNN, STAN, NARM, and a SIGMA-compatible reimplementation; further baselines are supplementary.

Allocation controls share identical expert ranks: equal mixing ($\beta = .5$), a learned OOF-global scalar, two independently learned short/long scalars ($L \leq 2$ versus longer; Amazon test rows use the longer side), and Eq. 3. We additionally test a fixed deterministic reassignment of the frozen β_q values across test queries (preserving their exact multiset), a 16-unit MLP, the 16 target-free diagnostics listed in the supplement, $k \in \{10, 20, 60\}$, a min–max normalized CombSUM perturbation under frozen β_q and $\beta = .5$. The primary three-feature linear gate and $\Delta = .10$ are fixed for all domains.

Paired per-query differences are averaged over matched seeds and bootstrapped over query IDs with 20,000 repetitions. Intervals condition on three fitted seeds; seed SD is reported separately. Experiments ran on an Apple M2 Pro with 32 GB memory; CEARF uses one CPU process and PASGR uses Metal/MPS.

Table 1. Compact allocation result on full-catalogue test queries. Values are matched-seed R@20 means. ΔU is dynamic minus OOF-global with a paired query-level 95% bootstrap interval; complete cutoffs and seed SD are in the supplement.

Domain	Memory	Neural	Equal .5	OOF global	OOF short/long	Dynamic	ΔU [95% CI]
Video	.11901	.12571	.14610	.14719	.14726	.14735	+.00022 [+.00009,+.00035]
Baby	.03879	.04873	.05555	.05671	.05655	.05669	+.00002 [-.00003,+.00006]
Diginetica	.49428	.44852	.51430	.51702	.51692	.51701	+.00002 [-.00015,+.00018]

Table 2. Full-catalogue test R@20. CEARF-N uses 3 seeds and text teachers (cached teacher documented as E5-small on Video; TF–IDF/SVD otherwise); comparator rows are ID-only, so this is system-level. Transition/V-SKNN/STAN are deterministic; complete baselines and cutoffs are supplementary. SIGMA-compatible is our reimplementation, not official SIGMA.

Method	Video	Baby	Diginetica
Transition-only	.04223	.00735	.40102
V-SKNN	.11936	.05038	.50506
STAN	.12115	.04944	.51502
NARM	.13705	.02990	**.53406**
SIGMA-compatible	.08644	.02473	.37219
CEARF-N	**.14735**	**.05669**	.51701

5 Results

5.1 RQ1–RQ2: endpoints, baselines, and allocation

Table 1 first establishes complementarity. On every domain, all fused allocations exceed both memory-only and neural-only at R@20. Equal mixing already gives a large endpoint gain; thus most of the improvement comes from heterogeneous evidence rather than gate complexity. The fused system evaluated with dynamic allocation exceeds the stronger endpoint by 17.2% on Video, 16.3% on Baby, and 4.6% on Diginetica. The stricter comparison is against the training-only global and short/long policies.

Table 2 is system-level context, not a metadata-matched architecture comparison. With the same cached teacher, NARM reaches .15387/.06628 R@20 on Video/Baby, exceeding CEARF-N; hence Amazon claims are allocation-orthogonal. Replacing PASGR by ID-only NARM inside the allocator gives .14854/.05001/.53939 R@20 on Video/Baby/Diginetica, closing Diginetica but losing Baby's metadata benefit. SIGMA-compatible underperforms transition-only on Diginetica despite selected epochs 7/9/10, so it is only context from our compatible reimplementation.

The final column of Table 1 isolates the query-wise correction. Dynamic allocation raises U on Video Games with a paired query-level interval above zero. It is not uniformly the best coarse-control row in U: fixed .5, short/long, and head/tail are slightly higher on Video, Baby, and Diginetica, respectively. Our claim is therefore narrower and more falsifiable: a five-scalar, source-disjoint residual exposes query-level expert advantage when it is stable while staying near the training-only global prior when it is not. Baby has a much smaller positive mean shift whose interval spans zero; Diginetica is likewise not detectably different from global. The gate can move the global coefficient by at most .10. Video improves most at early ranks; Baby's R@20 mean is slightly lower, so the correction is not uniform across cutoffs. On Video, dynamic-minus-global intervals are also above zero for R@6/R@10 (+.000278/+.000405) and nDCG@6/@10/@20 (+.000198/+.000238/+.000182), whereas R@20 is not; bounded correction changes early ordering more clearly than broad top-20 coverage.

The fixed same-multiset reassignment control holds the distribution of β_q fixed while changing only query assignment. Relative to a fixed target-free reassignment, the learned policy raises Video nDCG@20 by .000146 [.000082,.000210]; Diginetica has a supporting nDCG@10 difference of .000126 [.000033,.000221], while Baby intervals overlap zero. On Video, realized neural-minus-memory nDCG@20 also changes from -.01026 in beta deciles 1–3 to .00589 in deciles 8–10, a .01614 separation. Its standardized length/frequency coefficients are negative across all seeds and the tail coefficient is nonnegative, assigning more neural mass to shorter, lower-frequency contexts. Together, the fixed-reassignment and coefficient diagnostics are consistent with heterogeneous expert-advantage regimes, beyond merely changing the marginal weight distribution.

5.2 RQ3: parsimony and sensitivity controls

Capacity variants differ from the primary gate by only $O(10^{-4})$ in mean U: the all-feature linear gate loses on one matched seed in each domain, and Diginetica's best MLP is higher by only .00025. We keep the five-scalar form instead of selecting a larger gate post hoc. Relaxing the residual bound also does not explain the two null domains: $\Delta = .20$ gives the same Baby U (.04343) and essentially the same Diginetica U (.41329 versus .41332).

For each $k \in \{10, 20, 60\}$, the continuous prior and gate are refit on the same OOF training ranks; no setting is selected on validation or test. The primary $k = 20$ has the highest mean U on Video and Diginetica and lies within 1.1×10^{-4} of $k = 60$ on Baby. Complete sensitivity and fixed-allocation operator tables are in the supplement. Under frozen dynamic allocation, weighted RRF exceeds normalized CombSUM in R@20 by .00133/.00134/.00272 on Video/Baby/Diginetica; equal allocation reverses on Diginetica, indicating operator–allocation compatibility rather than unconditional RRF superiority.

5.3 RQ4: inference cost

For the timed seed-42 runs, the sum of separately measured warm medians for feature construction and gate evaluation is 1.171 μs/query on Video, 1.184 on Baby, and .774 on Diginetica. The jointly timed feature+gate+RRF pipeline costs 85.89, 87.57, and 80.60 μs/query, respectively. The supplement reports the expert-inclusive cross-run estimate. In a deterministic replay check, reconstructed β_q values differ from the original inference outputs by at most 5.96×10^{-8}, and the top-20 rankings are reproduced exactly. Expert retrieval dominates

Table 3. Cost–quality audit. Quality is recomputed from the frozen v2/evidence per-query arrays (the allocation tables report the separate dynamic-β lineage); tuning budgets are recorded from the audited grids; CEARF-N serving latency is recorded on Apple M2 Pro (warm).

Method	Budget	R@20 (V)	R@20 (B)	R@20 (D)	MRR@
CEARF-N	n/a	0.1460	0.0539	0.4903	
v2 bucketed router	3 router cells	–	–	0.4876	
GRU4Rec	≤12 ep.	0.1037	0.0450	0.4154	
NARM	≤12 ep.	0.1371	0.0299	0.4827	
v2 regime router	3 router cells	–	–	0.4875	
SASRec	≤12 ep.	0.0379	0.0316	0.0020	

CEARF-N serving latency (recorded): gate 1.171 μs, fusion 85.89 μs, expert 3.71 ms, total 3.796 ms/query (Video; similar on Baby

latency; the four-parameter residual gate is not the serving bottleneck. Table 3 completes the audit with quality recomputed from the frozen per-query arrays, the recorded tuning budget of every method, and MRR@20. The audited grids evaluate 8 cells for V-SKNN, 16 for STAN, and epoch-only selection (≤12, 5, and 10 epochs) for the neural baselines; the allocation-control family covers 15 policies. Gate and fusion together cost 87 μs/query on Video, 2.3% of the 3.8 ms total, so the allocator cannot be justified by serving-cost savings; its value must be argued from the allocation evidence alone.

5.4 RQ5: selection-procedure comparison

Table 4 compares six model-selection procedures directly, with their tuning budgets and regret against the test-side oracle over the fifteen-cell allocation-control family. Endpoint-only selection (P1) loses .02173/.00799/.02282 R@20 on Video/Baby/Diginetica, and per-seed recomputation from the frozen arrays confirms fusion beats the better endpoint in 3/3 seeds on both Amazon domains. The central audit result is that P2—a single training-learned global scalar with zero validation cells—already sits within 2.5×10^{-4} R@20 of the oracle on every domain. The query-conditioned gate (P6, the same fifteen-cell budget as the oracle search) improves the utility over P2 by $+.00022$ $[+.00009, +.00035]$ on Video but is statistically undetectable on Baby and Diginetica. Forbidding endpoint rejection (P5) costs nothing here because fusion dominates both endpoints on every domain and seed; the off-option is never selected, so "rejection completeness" carries no accuracy penalty in these regimes. We therefore report the simple-global result as the practical default and per-query conditioning as a detectable, domain-specific refinement rather than the primary source of accuracy.

5.5 RQ6: matched-information and expert-swap controls

Table 5 separates information conditions from mechanism conditions. With identical TF–IDF semantic initialization, NARM+sem exceeds CEARF-N full by +.0069 (Video) and +.0103 (Baby) R@20: metadata, not the fusion mechanism, drives most of the Amazon gain. The expert-swap control answers the converse question—whether fusion only rescues a weak expert. Inside the identical memory/gate/fusion stack, replacing PASGR with NARM raises the neural endpoint

Table 4. Selection-procedure comparison (test R@20; regret vs. test-side oracle over the 15-cell allocation-control family). P2 uses one training-learned scalar with zero validation cells.

		Test R@20			Regret vs. oracle		
Procedure	Budget	Video	Baby	Digi.	Video	Baby	Digi.
P1 endpoint-only	2	0.12571	0.04873	0.49428	0.02173	0.00799	0.02282
P2 global-β fusion	0	0.14719	0.05671	0.51702	0.00025	0.00001	0.00008
P3 regime router	2	0.14726	0.05655	0.51692	0.00018	0.00017	0.00018
P4 oracle (upper bnd.)	15	0.14744	0.05672	0.51710	0.00000	0.00000	0.00000
P5 forced neural incl.	4	0.14735	0.05671	0.51702	0.00009	0.00001	0.00008
P6 query-cond. gate	15	0.14735	0.05669	0.51701	0.00009	0.00003	0.00009

Table 5. Matched-information comparison and expert-swap generality (test R@20). Upper block: same-teacher conditions. Lower block: NARM replacing PASGR inside the identical memory/gate/fusion stack (3 seeds).

Condition	Init	Video	Baby
NARM (ID-only)	random	.13705	.02990
CEARF-N no-meta	random	.13208	.05055
NARM+sem	TF–IDF	**.15387**	**.06628**
CEARF-N full	TF–IDF	.14700	.05600
Neural-expert swap (PASGR $\rightarrow$ NARM), same memory+gate			
NARM-only	ID	.13763	.03880
gate fusion over NARM	ID	.14854	.05001
fusion gain		+.01091	+.01121

from .12571 to .13763 (Video) and from .44852 to .53178 (Diginetica), yet the same gate still adds +.01091, +.01121, and +.00761 R@20 over the swapped expert alone; on Diginetica fusion helps even though the replacement expert is the stronger single model. The gain is constructive at the query level: the recorded rescue/damage decomposition counts 4451/1766, 3889/1191, and 3076/1693 net-positive interventions on Video/Baby/Diginetica. The semantic conditions are recorded point estimates from the locked upstream audit; their per-query arrays were not retained, so no new paired interval is computed for them here.

6 Conclusion and Limitations

CEARF-N replaces validation-selected coarse mixing weights with a source-disjoint query-invariant prior and a bounded query-wise residual law. CEARF-N's main accuracy gain comes from expert complementarity; the residual asks the harder, narrower question of whether any source-disjoint query signal remains after a strong global prior. It does on Video's early ranks, while Baby/Diginetica are reported as boundary cases. The five-scalar allocator is cheap, bounded, score-scale invariant, and auditable. The procedure audit sharpens the message: a training-learned global scalar already recovers the fusion gain at zero validation budget, per-query conditioning is a small domain-specific addition, and the complementarity finding survives swapping in a stronger neural expert.

Three domains do not establish universal transfer, and Diginetica lacks original session IDs beyond expanded query rows for bootstrap clustering. Only

41.7%, 21.1%, and 82.5% of OOF calibration queries are actionable on Video, Baby, and Diginetica; the allocator is therefore trained conditionally on expert-union coverage, especially narrowly on Baby. The PASGR cell was frozen by an upstream validation study; our "training-only" claim applies specifically to allocation β, not to every expert hyperparameter. Semantic metadata makes external comparisons system-level; matched-teacher evidence exists only on Amazon. Finally, the bounded query-wise residual gain over global fusion is small and undetectable on two domains; larger untouched domains and temporal cross-fitting remain necessary. Complete implementation details, allocation controls, and statistical procedures are provided in the supplementary material.

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